Lecture 6

Homology and the Euler characteristic

Objectives. *Homology distinguishes the sphere from a point using the invariance theorem of §6.5. Both spaces have trivial fundamental group (Lecture 4). Homology compares closed chains with boundaries to count holes in each dimension. We compute it from boundary matrices for the torus and projective plane. Its alternating dimensions give the Euler characteristic. Binary cycle classes also describe closed strings in the toric code.*

By the end of this lecture you will be able to:

- Construct boundary matrices from ordered simplices.
- Compute cycles, boundaries and homology with specified coefficients.
- Prove the Euler characteristic formula for a finite complex.
- Identify the two independent binary string classes on a torus.

6.1 Simplices and their identifications

Definition 6.1 (Simplex). A *k-simplex* is the convex hull of $k+1$ affinely independent points, with a chosen ordering of these vertices. The standard simplex is

$$\Delta^k = \{(t_0, \dots, t_k) \in \mathbb{R}^{k+1} : t_i \geq 0,\ t_0 + \cdots + t_k = 1\}.$$

Deleting one vertex gives a *face*, with its inherited order.

A point, an interval and a triangle are simplices of dimensions zero, one and two. The order of the vertices orients an edge from the first vertex to the second. It also specifies the signs of higher-dimensional faces.

We identify the endpoints of one interval, as in Lecture 2, to obtain a circle. The domain endpoints v_0 and v_1 are distinct, but both map to the same vertex v. The interval interior maps injectively to the open edge. This circle has one vertex and one edge. The quotient map $q\colon \Delta^1 \to S^1$ sends the interval to the circle in the drawing, which also shows a three-edge decomposition.

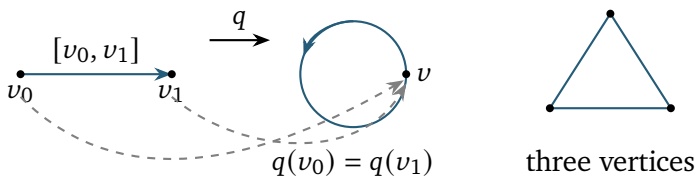


Figure 6.1: The interval's distinct domain endpoints both map to v. The triangle outline is a simplicial decomposition of the same circle.

Definition 6.2 (Δ-complex). A Δ-*complex structure* on X is a collection of continuous maps $\sigma\colon \Delta^k \to X$, called its simplices, with the following properties. Each map is injective on the interior of its domain. These interiors have disjoint images and together cover X. The restriction to each face is a designated lower-dimensional simplex, using the affine identification preserving the vertex order. The topology is specified by these maps: a subset $A \subseteq X$ is closed exactly when $\sigma^{-1}(A)$ is closed for every simplex σ.

The closed-set condition gives the quotient topology of Lecture 2. Thus the space is obtained by attaching the faces in the specified way.

Different vertices of one domain simplex may have the same image in X. An edge can therefore become a loop. By contrast, in a *simplicial complex* each simplex has distinct vertices in the space and any two simplices intersect in a common face or not at all. The torus model below has one vertex and two triangles. It is a Δ-complex, not a simplicial complex. Vertex names such as $[v_0, v_1, v_2]$ refer to ordered domain vertices even when some images coincide.

6.2 Chains and the boundary map

Fix a *coefficient ring* R. Our default is $R = \mathbb{Z}$. The alternatives used here are $\mathbb{Z}_2$, the field with arithmetic modulo two, and the real field $\mathbb{R}$. Integers form a ring, not a field.

Definition 6.3 (Chains). A k-*chain* is a finite formal sum of the k-simplices with coefficients in R. These sums form the *chain group* $C_k(X; R)$.

Chains are added and multiplied by coefficients componentwise. Every chain has a unique expression in the simplex generators. When R is a field, they form a vector-space basis. Write $C_{-1} = 0$.

Definition 6.4 (Boundary map). The *boundary map* $\partial_k\colon C_k(X;R) \to C_{k-1}(X;R)$ is the linear map defined on a simplex by alternating deletion:

$$\partial_k[v_0, \dots, v_k] = \sum_{i=0}^{k} (-1)^i [v_0, \dots, \widehat{v_i}, \dots, v_k] \quad (k \geq 1), \qquad \partial_0 = 0.$$

The hat means that the vertex is omitted. Each face is read as the designated simplex of the complex. This ∂ acts on chains. It is not a coordinate derivative.

For an ordered triangle, the formula is

$$\partial[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1].$$

The chain records the circuit from v_0 to v_1, then to v_2, and back to v_0. The negative sign reverses the contribution of the middle edge. An integer chain such as $2a - b$ records two copies of edge a and one oppositely oriented copy of edge b. A chain need not form a single continuous path. Over $\mathbb{Z}_2$ the negative sign equals a positive sign. Repeated occurrences of an edge still have to be added in the chosen ring. The drawing separates the ordered edges from their boundary circuit.

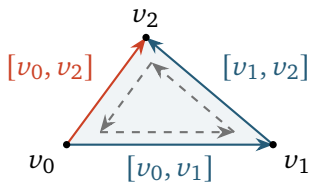


Figure 6.2: The ordered edges point from lower to higher vertex index. The dashed boundary circuit traverses the middle edge with a negative coefficient.

Theorem 6.5 (The boundary of a boundary)**.** *For every $k \geq 1$, the composition $\partial_{k-1}\partial_k$ is zero.*

Proof. For $k = 1$ the target boundary is $\partial_0 = 0$. For $k \geq 2$, fix indices $i < j$ in an ordered k-simplex. Deleting v_i first moves v_j to position $j - 1$. The resulting face has coefficient $(-1)^i(-1)^{j-1}$. Deleting v_j first leaves v_i in position i and gives coefficient $(-1)^j(-1)^i$. These are opposite signs. Every face with two deleted vertices occurs in exactly this pair. The pairs cancel before the face identifications, so they also cancel after those identifications. Linearity proves the assertion for every chain. ∎

For the triangle, applying the boundary once more gives

$$(v_2 - v_1) - (v_2 - v_0) + (v_1 - v_0) = 0.$$

Here a vertex name denotes its generator in C_0. For a one-cycle, zero boundary means that the signed endpoint contributions cancel. The boundary of a triangle has no remaining endpoints.

Definition 6.6 (Chain complex)**.** A sequence of groups linked by homomorphisms whose consecutive compositions vanish is called a *chain complex*.

6.3 Cycles, boundaries and coefficients

Definition 6.7 (Cycles)**.** A *cycle* is a chain with zero boundary. Its group is

$$Z_k(X; R) = \ker \partial_k.$$

Definition 6.8 (Boundaries)**.** A *boundary* is a chain that is the boundary of a chain one dimension higher. Its group is

$$B_k(X; R) = \operatorname{im} \partial_{k+1}.$$

The boundary-of-boundary theorem gives $B_k \subseteq Z_k$.

Definition 6.9 (Homology)**.** The *homology group* in degree k is

$$H_k(X; R) = Z_k(X; R)/B_k(X; R).$$

Two cycles are *homologous* when their difference is a boundary.

Homology records which cycles remain after this identification. The symbols Z_k and B_k denote groups, distinct from the coefficient ring $\mathbb{Z}$ and the braid group B_n of Lecture 5.

For example, take the boundary of a triangle, with no filled face. The sum z of its cyclically directed edges is a cycle. A general cycle has equal coefficients on those three directed edges, since its boundary vanishes at each vertex. There are no two-dimensional chains, so this cycle generates $H_1 \cong R$. Filling the triangle makes that same sum a boundary and gives $H_1 = 0$.

Since $\partial_0 = 0$, every zero-chain is a cycle. The boundary of an edge from v_i to v_j is $v_j - v_i$, so the endpoint classes become equal in H_0. Edge paths therefore identify all vertices in one component.

In a finite Δ-complex, these edge-path components are exactly the path components: each simplex joins its vertices by edges, and its points by paths to those vertices. Distinct edge-path components are disjoint finite unions of closed simplices, so they are also open.

Summing vertex coefficients separately in each component shows that there is no relation between different components. Thus H_0 has one independent generator for each path component. Over $\mathbb{Z}$ this is a free abelian group. Over a field it is a vector space.

The one-edge circle has zero edge boundary and no two-simplices, so its zeroth and first homology groups are both R. The triangle outline has the same answer. Filling changes its first homology but not its component count, as the following panels show.

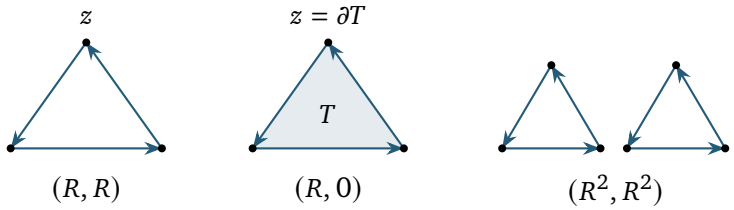


Figure 6.3: With $R \in \{\mathbb{Z}, \mathbb{Z}_2, \mathbb{R}\}$, the panels show $(H_0, H_1) = (R, R)$, $(R, 0)$ and (R^2, R^2). In the filled panel the cycle is the boundary of the face T.

One dimension higher, the boundary of a tetrahedron is a model of S^2. Its ordered vertices are $0, 1, 2, 3$. The oriented surface chain is

$$z_2 = [123] - [023] + [013] - [012].$$

The boundary-of-boundary theorem gives $\partial z_2 = 0$. Cancellation on shared edges forces every two-cycle to be a multiple of z_2. There are no three-simplices, so $H_2(S^2; R) \cong R$. Every one-cycle is a boundary: subtract boundaries of faces $[0ij]$ to remove all edges not incident to 0. The remaining cycle uses only edges from 0, whose other endpoints force their coefficients to vanish. We obtain

$$H_0(S^2; R) \cong R, \qquad H_1(S^2; R) = 0, \qquad H_2(S^2; R) \cong R.$$

For a point, only H_0 is nonzero. By the invariance theorem of §6.5, this distinguishes the sphere from a point even though their fundamental groups agree. Filling the tetrahedron makes z_2 a boundary. The drawing shows the surface before and after filling.

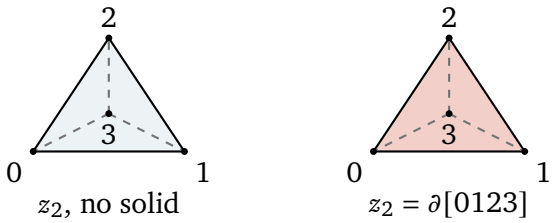


Figure 6.4: The same tetrahedral surface cycle z_2 is not a boundary in the surface complex, but is the boundary of $[0123]$ in the solid complex.

Homologous cycles need not coincide as chains. For example, triangulate a square annulus with every triangle counter-clockwise. Its outer and inner boundary circuits c_1 and c_0 can both be directed counter-clockwise. The sum B of its oriented triangles then satisfies

$$\partial B = c_1 - c_0.$$

The inner induced direction is clockwise, accounting for the minus sign. Thus c_1 and c_0 represent the same homology class. The shaded annulus displays this boundary relation.

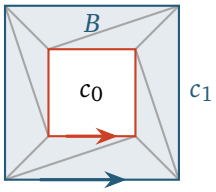


Figure 6.5: Both marked circuits run counter-clockwise. The oriented annulus has boundary $c_1 - c_0$, so their difference is a boundary.

For a finite complex, order the simplices of each dimension. The boundary matrix D_k has a column for each source k-simplex and a row for each target $(k-1)$-simplex. Its entries are the signed face coefficients. Write c_k for the number of k-simplices and r_k for the rank of D_k over a specified field. Rank–nullity and the dimension formula for a quotient, recalled from *Linear Algebra for Physics*, give

$$\dim H_k(X; R) = (c_k - r_k) - r_{k+1} \qquad \text{when } R \text{ is a field.}$$

Over $\mathbb{Z}$, integer relations determine an abelian-group quotient instead. Real row reduction alone does not determine that group.

Definition 6.10 (Torsion)**.** An element is a *torsion element* if it is nonzero and some positive integer multiple is zero.

For an algebraic example, take the chain complex with $C_2 = C_1 = \mathbb{Z}$, boundary ∂_2 given by multiplication by two, and $\partial_1 = 0$. Its first homology is $\mathbb{Z}/2\mathbb{Z} = \mathbb{Z}_2$. With real coefficients, multiplication by two is onto, so the first homology is zero. With $\mathbb{Z}_2$ coefficients it is the zero map, so the first homology is $\mathbb{Z}_2$. This illustrates why coefficients are part of the answer.

Definition 6.11 (Betti number)**.** For a finite complex, the *kth Betti number* means $b_k = \dim_{\mathbb{R}} H_k(X; \mathbb{R})$.

For $k \geq 1$ this counts independent real cycle classes modulo boundaries, often called k-dimensional holes. The number b_0 counts path components. Real coefficients discard torsion information.

6.4 The Euler characteristic

The alternating simplex count can be computed without reducing boundary matrices. We relate it to homology to explain why a different decomposition gives the same count.

Definition 6.12 (Euler characteristic)**.** For a finite Δ-complex of dimension d, its *Euler characteristic* is the alternating simplex count

$$\chi(X) = \sum_{k=0}^{d} (-1)^k c_k.$$

For a triangulated surface this is vertices minus edges plus triangles. The symbol χ denotes a topological count here, not the character of Lecture 5.

Theorem 6.13 (Euler characteristic and homology)**.** *For a finite Δ-complex and any field R, $\chi(X) = \sum_{k=0}^{d} (-1)^k \dim_R H_k(X; R)$. In particular, $\chi(X) = \sum_k (-1)^k b_k$.*

Proof. Set $r_0 = r_{d+1} = 0$. The dimension formula for homology gives $c_k = \dim H_k + r_k + r_{k+1}$. On summing with alternating signs, the rank terms cancel:

$$\sum_{k=0}^{d} (-1)^k c_k = \sum_{k=0}^{d} (-1)^k \dim H_k + \sum_{k=0}^{d} (-1)^k r_k + \sum_{k=0}^{d} (-1)^k r_{k+1} = \sum_{k=0}^{d} (-1)^k \dim H_k.$$

Indeed, the coefficient of each r_j with $1 \le j \le d$ is $(-1)^j + (-1)^{j-1} = 0$. The endpoint ranks vanish. ∎

For a single filled triangle, the counts are 3, 3 and 1, giving $\chi = 1$. Its homology has one degree-zero generator and no higher generators, which gives the same alternating sum.

6.5 Invariance and closed surfaces

The calculation uses a decomposition. Its independence of that choice requires a theorem, rather than a comparison of a few examples.

Theorem 6.14 (Homology invariance · *cited*)**.** *The homology of a Δ-complex agrees with its singular homology. Homotopy equivalences induce isomorphisms of these groups, with the coefficient ring fixed. Consequently the Euler characteristic of a finite Δ-complex is invariant under homotopy equivalence.*

The comparison is Hatcher, Theorem 2.27. Homotopy invariance is Corollary 2.11. Coefficients are treated in §2.2, pp. 153–155. Singular homology uses all continuous maps $\Delta^k \to X$ as generators, without the interior-injectivity requirement. That construction applies without choosing a decomposition. The Euler assertion follows from the theorem above, for example with real coefficients.

Definition 6.15 (Triangulation)**.** A *triangulation* is a presentation of a surface by triangles glued along edges.

Definition 6.16 (Orientability)**.** A triangulated surface is *orientable* if its triangles can be oriented so that their induced directions on every shared edge are opposite.

We use as part of the cited surface classification below that every closed surface admits such a presentation and that this condition is independent of the choice. The torus has this property. The projective plane fails the sign test in §6.6.

The Möbius band of Lecture 2 illustrates the obstruction. After rescaling the transverse coordinate to $t \in [-1, 1]$, its rectangle model identifies $(1, t)$ with $(0, -t)$. Orienting adjacent triangles consistently in the rectangle makes the induced directions opposite on their interior edges. At the seam, the reversal of t makes the induced directions agree instead. Reversing every triangle changes both seam directions and cannot fix this. The drawing compares an interior edge with this seam.

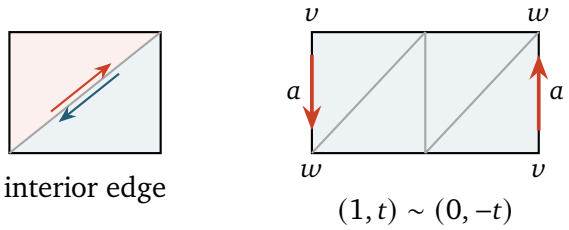


Figure 6.6: Counter-clockwise face orientations cancel on an interior edge. The seam $(1, t) \sim (0, -t)$ identifies the two marked boundary directions, so they agree on the Möbius band.

Theorem 6.17 (Closed surfaces · *cited*)**.** *Closed connected surfaces are classified, up to homeomorphism, by their orientability and Euler characteristic. For the orientable surface Σ_g of genus $g \geq 0$, the Euler characteristic is $2 - 2g$, where $\Sigma_0 = S^2$.*

This is the classification theorem of Kosniowski, Theorem 11.3, together with the Euler counts in Hatcher, §2.2, pp. 146–147. For $g \geq 1$, the polygon model of Lecture 5 has one vertex, $2g$ edges and one face. Using such polygonal cells instead of triangles gives the same Euler characteristic by Hatcher, Theorem 2.44. Thus its count is $1 - 2g + 1$.

There is also a direct connection to Lecture 5. For any path-connected space, $H_1(X; \mathbb{Z})$ is the *abelianization* of $\pi_1(X)$: the quotient by the normal subgroup generated by all commutators $aba^{-1}b^{-1}$. This result is cited from Hatcher, Theorem 2A.1. Homology therefore forgets the order information in noncommuting loop classes. It need not recover the fundamental group.

6.6 Boundary matrices for two surfaces

We compute the torus over $\mathbb{Z}$ and then $\mathbb{Z}_2$. Take a square with corners P, Q, R', S counter-clockwise from the bottom left. The prime distinguishes the corner R' from the coefficient ring R. Pair the horizontal edges to the right and the vertical edges upward. All corners become one vertex v. Call the horizontal edge a, the vertical edge b and the diagonal $P \to R'$ edge c. The ordered triangles are $T_0 = [P, Q, R']$ and $T_1 = [P, S, R']$.

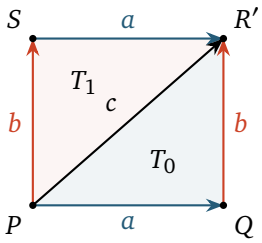


Figure 6.7: The torus has one vertex, three edges and two triangles. The upper triangle's vertex order has the opposite planar orientation to the lower one's.

Example 6.18 (The torus)**.** Both triangle boundaries are $a + b - c$. Each edge has zero boundary, since both endpoints become v. In the bases (a, b, c) and (T_0, T_1), the boundary matrices over $\mathbb{Z}$ are

$$D_1 = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}.$$

$$D_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \end{pmatrix}.$$

Every one-chain is a cycle. The boundaries are integer multiples of $a + b - c$. The list $(a, b, a + b - c)$ is an integer basis of C_1: its change-of-basis determinant is -1, so its inverse has integer entries. Quotienting by its last generator leaves a and b free. The kernel of D_2 consists of pairs $(m, -m)$, and there are no three-chains. The zero-dimensional quotient retains the single vertex. We have therefore obtained

$$H_0(T^2; \mathbb{Z}) = \mathbb{Z}, \qquad H_1(T^2; \mathbb{Z}) = \mathbb{Z}^2, \qquad H_2(T^2; \mathbb{Z}) = \mathbb{Z}.$$

The top generator is $T_0 - T_1$. Its signs give both triangles the same planar orientation, and their edge contributions cancel.

With $\mathbb{Z}_2$ coefficients, every entry -1 becomes 1. The ranks of D_1 and D_2 are zero and one. The homology dimensions are therefore 1, $3 - 0 - 1 = 2$ and $2 - 1 = 1$. The classes of a and b still generate H_1, and the top cycle is $T_0 + T_1$. Both coefficient choices give $\chi(T^2) = 1 - 3 + 2 = 0$.

For the projective plane, pair opposite square edges with opposite arrows. The corners P and R' become v, while Q and S become w. The edge a runs from P to S and from R' to Q. The edge b runs from P to Q and from R' to S. The diagonal c runs from P to R'.

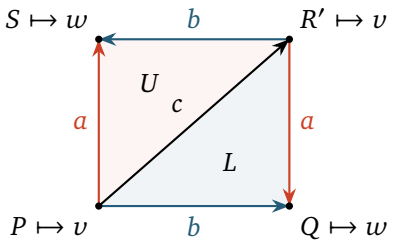


Figure 6.8: The diagonal c is a loop at v in this two-vertex model of the projective plane. Both boundary-edge generators join v to w.

Example 6.19 (The projective plane over $\mathbb{Z}_2$)**.** Use the vertex basis (v, w), edge basis (a, b, c) and ordered triangles $U = [P, R', S]$ and $L = [P, R', Q]$. Work over $\mathbb{Z}_2$ throughout. Both edges a and b have boundary $v+w$, and c has zero boundary. Both triangles have boundary $a+b+c$. Thus

$$D_1 = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}.$$

$$D_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Both matrices have rank one. The kernel of D_1 is generated by $a + b$ and c. Its boundary subspace is generated by $a + b + c$, so the quotient is one-dimensional, represented by c. The kernel of D_2 is generated by $U + L$. The image of D_1 identifies the two vertex classes in H_0. Consequently $H_k(\mathbb{R}\mathrm{P}^2; \mathbb{Z}_2) \cong \mathbb{Z}_2$ for $k = 0, 1, 2$, and $\chi(\mathbb{R}\mathrm{P}^2) = 2 - 3 + 2 = 1 = 1 - 1 + 1$. These are binary homology groups. They do not assert the same answer with integer or real coefficients.

We now use integer coefficients with the same arrows and ordered triangles. Their boundaries are

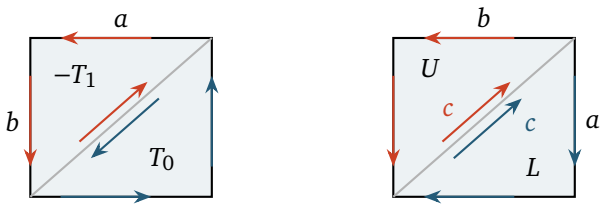
$$\partial a = \partial b = w - v, \qquad \partial c = 0, \qquad \partial U = b - a + c, \qquad \partial L = a - b + c.$$

Put $x = a - b$. The integer one-cycles have basis (x, c). The boundaries are generated by $c - x$ and $c + x$. The first relation identifies x with c. The second then becomes $2c = 0$. The class of c is nonzero: solving $m(c - x) + n(c + x) = c$ forces $m = n$ and $2m = 1$, which has no integer solution. Thus

$$H_0(\mathbb{R}\mathrm{P}^2; \mathbb{Z}) \cong \mathbb{Z}, \qquad H_1(\mathbb{R}\mathrm{P}^2; \mathbb{Z}) \cong \mathbb{Z}_2, \qquad H_2(\mathbb{R}\mathrm{P}^2; \mathbb{Z}) = 0.$$

For the last assertion, a two-cycle $mU + nL$ would require $m = n$ at a and $m = -n$ at c, hence $m = n = 0$. The relation $\partial(U + L) = 2c$ gives a geometric example of torsion. The diagonal joins antipodal boundary points P and R' in the square model. We identify that square with a hemisphere, as in Lecture 2. The corresponding lift to S^2 has distinct antipodal endpoints. By the covering argument of Lecture 4, the diagonal represents the nontrivial fundamental-group class of $\mathbb{R}\mathrm{P}^2$. The fundamental group $\mathbb{Z}_2$ is already abelian, agreeing with the abelianization result in §6.5. Over $\mathbb{R}$, the relations $x = c$ and $2c = 0$ instead give $c = 0$. The real Betti numbers are therefore $(b_0, b_1, b_2) = (1, 0, 0)$.

For an orientation test, the coefficients m and n must each be $+1$ or -1. The conditions $m = n$ and $m = -n$ cannot both hold. Over $\mathbb{Z}_2$, their distinction disappears and $U + L$ is a top cycle. The comparison with the torus shows precisely which edge contributions fail.



$\partial(T_0 - T_1) = 0$ $\quad$ $\partial(U + L) = 2c$, $2c = 0$ over $\mathbb{Z}_2$

Figure 6.9: The torus cycle $T_0 - T_1$ cancels every edge contribution. For $U + L$ on $\mathbb{R}P^2$, the a and b contributions cancel but the two copies of c add to $2c$, which vanishes only after reduction modulo two.

The classification theorem and these computations give the following comparison.

Surface	Orientable	χ
S^2	yes	2
T^2	yes	0
Σ_2	yes	−2
$\mathbb{R}P^2$	no	1
Klein bottle	no	0

The Klein bottle is the square quotient with horizontal edges paired in the same direction and vertical edges paired with reversal. Its reversed seam fails the same orientation test as the Möbius band. Its polygonal cells give $\chi = 1 - 2 + 1 = 0$ by the cited cell-count theorem. Thus the Klein bottle and torus have the same Euler characteristic. Their different orientability explains why the classification needs both data.

Self-check. Compute the Euler characteristic of the sphere from the boundary of a tetrahedron and from the boundary of an octahedron. Confirm the tetrahedral answer from the homology of its boundary.

Answer. The tetrahedron has four vertices. Every pair is an edge, giving $\binom{4}{2} = 6$ edges, and every triple is a face, giving four triangles. The count is $4 - 6 + 4 = 2$. The octahedron has vertices at the positive and negative coordinate-axis unit vectors. Every vertex joins the four vertices other than itself and its opposite, giving $6 \cdot 4/2 = 12$ edges. A face chooses one endpoint from each axis, giving $2^3 = 8$ triangles. Its count is $6 - 12 + 8 = 2$, in agreement with the tetrahedron. The tetrahedral boundary has real Betti numbers $(1, 0, 1)$, so its homological alternating sum is also $1 - 0 + 1 = 2$.

Self-check. Consider the chain complex $C_2 = \mathbb{Z}$, $C_1 = \mathbb{Z}^3$ and $C_0 = \mathbb{Z}^3$, with bases (T), (a, b, c) and (v_0, v_1, v_2). Its only nonzero boundary matrices are

$$D_1 = \begin{pmatrix} -1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad D_2 = \begin{pmatrix} 5 \\ 5 \\ -5 \end{pmatrix}.$$

Compute $\partial(2a - b + 3c)$ and H_1 over $\mathbb{Z}$ and over $\mathbb{R}$.

Answer. The chain has boundary $-5v_0+3v_1+2v_2$, so it is not a cycle. Solving $D_1(u, v, w)^{\mathsf{T}} = 0$ gives $u = v = -w$. Hence the cycles are generated by $z = a + b - c$, and $\partial T = 5z$. The quotient is $H_1 \cong \mathbb{Z}/5\mathbb{Z}$ over $\mathbb{Z}$. Over $\mathbb{R}$, multiplication by five is onto, so $H_1 = 0$. The same real rank does not determine the integer quotient.

6.7 Binary cycles and the toric code

We use homology to distinguish strings that leave stored information unchanged from strings that can change it. The model is Kitaev's toric code (arXiv:quant-ph/9707021v1, §1, equations (1)–(2)).

Physical input. In the ideal finite-qubit model, normalized vectors in a complex tensor product describe pure states. A qubit has orthonormal basis $|0\rangle$ and $|1\rangle$. Measuring a Hermitian operator on an eigenstate returns its eigenvalue. We interpret the joint +1 eigenspace defined below as the states used to store quantum information, following Kitaev, §1.

The phase-flip operator Z and bit-flip operator X are given by

$$Z|0\rangle = |0\rangle, \qquad Z|1\rangle = -|1\rangle, \qquad X|0\rangle = |1\rangle, \qquad X|1\rangle = |0\rangle.$$

Thus $X^2 = Z^2 = 1$ and $XZ = -ZX$, as follows by applying them to both basis vectors. The operator Z is distinct from the cycle group Z_k.

Definition 6.20 (Toric code)**.** Take an $N \times N$ square lattice with opposite sides identified, where $N \geq 3$. Put one qubit on each edge. The joint state space is

$$\mathcal{H} = (\mathbb{C}^2)^{\otimes 2N^2}.$$

The notation X_e and Z_e means that the operator acts on edge e and as the identity on every other factor. For each vertex v and each square face, called a *plaquette* p, define the *stabilizer operators*

$$A_v = \prod_{e \ni v} X_e, \qquad B_p = \prod_{e \in \partial p} Z_e.$$

The *toric code subspace* is their joint +1 eigenspace

$$C = \{\psi \in \mathcal{H} : A_v\psi = \psi \text{ for every } v,\ B_p\psi = \psi \text{ for every } p\}.$$

All stars A_v commute with one another. All plaquettes B_p also commute. A star and a plaquette share zero or two edges. Every shared edge contributes one minus sign from $XZ = -ZX$, so their even overlap makes $A_vB_p = B_pA_v$.

Definition 6.21 (Z-strings)**.** For a binary one-chain $c = \sum_e c_e e$, the *Z-string* is

$$Z(c) = \prod_e Z_e^{c_e}, \qquad c_e \in \mathbb{Z}_2.$$

Definition 6.22 (Logical operators)**.** A *logical operator* preserves the code subspace and acts on the states stored there.

Adding binary chains toggles selected edges, since $Z_e^2 = 1$. At a vertex v, commuting A_v past the string gives

$$A_vZ(c) = (-1)^{(\partial c)_v} Z(c)A_v.$$

Here $(\partial c)_v$ is represented by 0 or 1. For a code state ψ, the state $Z(c)\psi$ has star eigenvalue -1 exactly at vertices with odd incident string parity. The binary record of violated conditions is the *syndrome* of a Z-type error. It is precisely ∂c. Along a simple open string only the two ends violate star conditions. Every Z-string commutes with every B_p. Consequently strings with $\partial c = 0$ preserve C.

A face-boundary string is $Z(\partial p) = B_p$, which acts as the identity on C. Products of these operators show that any boundary acts as the identity. Homologous closed strings therefore act alike on the code. The drawing separates the local stabilizers from open and closed strings.

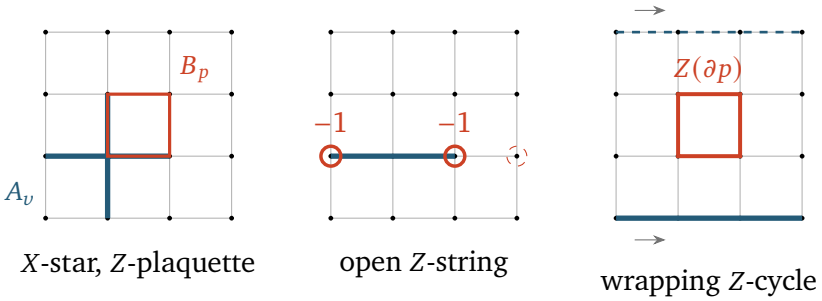


Figure 6.10: The code conditions are $A_v = B_p = +1$, with dashed marks repeating periodic copies. An open Z-string violates stars at its two ends, a plaquette boundary acts identically on the code, and a wrapping string has no endpoint syndrome.

The periodic cycles and a plaquette change can also be drawn on the full fundamental square.

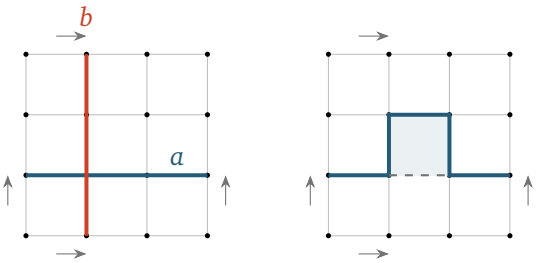


Figure 6.11: On the periodic square with opposite sides identified, the first panel shows the two wrapping cycles. The second panel adds one plaquette boundary, with the dashed edge removed.

The lattice uses square faces, whereas our homology was defined using simplices. We compare the two quotients by adding diagonals. We draw every square's diagonal from lower left to upper right and order its triangles as in the torus example. A triangle boundary replaces each added diagonal in a cycle by its other two edges. Conversely, each diagonal occurs in exactly the two triangles of its square. A binary sum of triangle boundaries with no diagonal term therefore uses the two triangles of each square with equal coefficients. The two triangle boundaries add to the square boundary, so the sum is a sum of square boundaries. This proves that square-lattice cycles modulo square boundaries agree with the homology of this triangulation. By the cited Theorem 6.14, its homology agrees with that of the two-triangle torus in Example 6.18.

Example 6.23 (Toric-code string classes · *cited; two-triangle homology proved*)**.** The binary string classes form $H_1(T^2;\mathbb{Z}_2) \cong \mathbb{Z}_2^2$. Representatives wind once in the horizontal and vertical directions. Every class is a sum of a subset of these two generator classes, including the empty sum. There are therefore four classes. The square-to-triangle comparison was proved above. Its comparison with the two-triangle torus uses the cited homology-invariance

theorem. The identification of these lattice representatives with logical Z operators is Kitaev's construction, with the parity calculation given here.

We can also derive the dimension of C. There are N^2 stars and N^2 plaquettes, with relations

$$\prod_v A_v = 1, \qquad \prod_p B_p = 1.$$

Each edge occurs twice in each product. These are the only relations within each family. A product of selected stars is the identity only if the selection agrees at both ends of every edge. The vertex graph is connected, so either every vertex or no vertex is selected.

The same argument on the connected dual graph, whose vertices are faces and whose edges join neighbouring faces, applies to plaquettes. A product of X operators and a product of Z operators can multiply to a scalar identity only when both supports are empty. Hence there are $r = 2N^2 - 2$ independent conditions, with no relation giving -1.

For independent commuting stabilizers $S_1, \dots, S_r$, the projector onto their joint $+1$ eigenspace is

$$P = 2^{-r} \prod_{j=1}^{r} (1 + S_j).$$

Each factor $(1 + S_j)/2$ is the projector onto its $+1$ eigenspace. Every nonidentity product in the expansion has, on some edge, a factor X, Z or XZ, up to a sign. Each of these has trace zero. The trace of a tensor product is the product of its factor traces. Only the identity contributes to the trace. Thus

$$\dim C = \operatorname{tr} P = 2^{2N^2 - r} = 4.$$

This is the state-space dimension of two logical qubits. The complementary logical operators come from X-strings on the dual lattice, as in Kitaev, §1. A closed dual string applies X on the primal edges it crosses. It meets each plaquette boundary evenly, so it preserves C. For each of the horizontal and vertical wrapping Z-strings, we choose a closed dual string crossing it once. The two operators anticommute, so the wrapping Z-string cannot act as the identity on C.

The distinction matters for errors in this ideal model. A Z-type error supported on fewer than N edges either has a nonzero syndrome or is a boundary and acts identically on the code. For the length bound, decompose a binary cycle into closed edge walks. A walk with nonzero winding lifts to a lattice walk whose endpoint displacement has a coordinate of magnitude at least N. It therefore needs at least N edges. A cycle of smaller total length has no wrapping component and is a boundary. This is the Z-string part of Kitaev's error-detection argument in §1. The interpretation as a quantum code and the complementary operators remain those of the cited construction.

Self-check. State in one sentence what $H_k(X; R)$ is and why it was introduced. State in a second sentence what the toric code subspace is and why it was introduced.

Answer. The group $H_k(X; R)$ consists of k-cycles modulo boundaries, so it distinguishes closed chains that cannot be filled within X. The toric code subspace is the joint $+1$ eigenspace of all A_v and B_p, introduced to store states on which boundary strings act identically while nontrivial wrapping strings can act nontrivially.

Summary. The boundary map has square zero. Homology is the quotient of cycles by boundaries, with the coefficient ring specified. Over a field, boundary-matrix ranks give its dimensions and their alternating sum is the Euler characteristic. Invariance and surface classification are cited. The torus has two independent binary one-cycle classes, which describe the two logical Z-string classes in the toric code. Lecture 7 adds smooth coordinates to spaces so that calculus can be defined on them.

Reading. Hatcher, *Algebraic Topology*, §2.1, pp. 102–107, Theorems 2.27 and 2A.1, and §2.2, pp. 146–147; Nakahara, *Geometry, Topology and Physics*, §3.3; Kosniowski, *A First Course in Algebraic Topology*, Theorem 11.3 and Chapter 29; Kitaev, *Fault-tolerant quantum computation by anyons* (2003); section numbers here refer to arXiv:quant-ph/9707021v1, §1.